Creating the spinosaurus animatronic—among the largest animatronic creations ever built for a motion picture—for the movie, Jurassic Park 3

Photo Courtesy: *Stan Winston Studio*

First, an artist creates a preliminary sketch of the creature, with the help of a specialist, say a palaeontologist, as for a dinosaur. Then a maquette (miniature model) is built which is followed by a full-size sculpture. A mould is created from the sculpture and the body is cast. Then the animatronic components are put together and tested for bugs.

Building the various components usually takes the longest time. Most of the creatures developed at the California-based Stan Winston Studio (SWS)—creators of special effects in movies such as *A.I. Artificial Intelligence*, *Jurassic Park*, *What Lies Beneath*, *Alien*, *The Terminator* and *The Sixth Sense*—require parts that are not easily available. For the Spinosaurus, SWS is said to have built almost everything themselves, dividing their work into three main categories:

- **Mechanical:** Nearly all of the mechanical systems used in the Spinosaurus are hydraulic. Another group developed the electronic control systems needed to operate the animatronic. Almost all the movements of the Spinosaurus are manipulated by specialised remote-control systems known as telemetry devices.
- **Structural:** All electronic and mechanical components need something to attach to and control and the skin must have a frame to maintain its shape. This is done by building a plastic and steel frame.
- **Surface:** The skin of the Spinosaurus is made of foam rubber. This is a special light and spongy rubber that is made by mixing air with liquid latex rubber and then curing (hardening) it.

But I just want my home video

The list of special effects is endless. This may inspire you to make a small movie yourself. If you own a digital camcorder, all you need to do is point and shoot to create a home video. Once you have all the raw footage, you can convert the scenes into a polished, professional production. Even George Lucas shot a large part of the *Attack of the Clones* with Sony HDW-F900 HDCAM camcorders outfitted with high-end Panavision lenses and saved a lot of cash and film.

If you plan to use your desktop computer for video editing, it should have a FireWire (IEEE 1394 to connect different pieces of equipment) port to connect to the camera. These support the DV editing format. We now have a few camcorders available with USB 2.0 ports. These are faster than FireWire and support the DVD format but not DV. We recommend a fast CPU and lots of memory since rendering and writing can consume a lot of power and disk space. Then you need video-editing software.

If you use Adobe Premiere, for instance, the process will create huge AVI or MOV files on your hard disk. You then need to choose the shots you need and arrange them, making a timeline. You line the shots up in sequential order and then play them as a sequence.

And how about my digital camera? First, the audio experience is missing (there are a few digital cameras that have audio capabilities too). Most digital camcorders capture about 30 fps (NTSC) or 25 fps (PAL), while most digital still cameras capture only 15 fps (there are exceptions such as the Sony F717 digital). Finally, digital cam recorders are capable of capturing at least an hour of motion and sound while most digital cameras can shoot only about 15-30 seconds. But given the pace at which technology moves, it's only a matter of time till we can have a full motion movie on a digicam.

Video-editing software

There are more than 150 video-editing software available for use such as Adobe Premiere 6.5. While working with Real-Time Preview, you can use the Adobe Title Designer, MPEG-2 export (Windows only), DVD authoring (Windows only) and audio tools to create video productions. The titling feature, Adobe Title Designer, comes with 90 PostScript fonts. Mac users obviously can't take full advantage of the Windows-only features. Premiere ships with good documentation, including a well-written user guide, online help and an excellent training CD-ROM from Total Training. It comes bundled with a third-party DVD-authoring program, Sonic Solutions DVDit LE.

Final Cut Pro is the only nonlinear editor that supports DV, SD and HD editing formats and analog video. It gives several options for managing your captured footage and includes a suite of powerful tools for adding cool effects and titles, and for compositing. You can get a real-time effects preview with the built-in G4 real-time effects architecture or use third-party PCI cards for real-time editing with render-free playback at SD and HD. You can create 2D and 3D animated titles, video cubes and page curls with tools from BorisFX, CGM, and third party After Effects plugins. You can work around the differences between cameras, lighting situations and even capture devices with the high-precision video calibration tools. Final Cut Pro 3 lets you output your projects for the Web (with compression), DVD or broadcast in DV, SD or HD format. It provides simultaneous video output to a computer monitor, an NTSC or PAL TV monitor, a VCR or a camera. But at nearly Rs 50,000, it is relatively over-priced for the home user.